Chapter 7

The “Super” in Superstrings

W

hen the success of Eddington’s 1919 expedition to measure Einstein’s prediction of the bending of starlight by the sun had been established, the Dutch physicist Hendrik Lorentz sent Einstein a telegram informing him of the good news. As word of the telegram’s confirmation of general relativity spread, a student asked Einstein about what he would have thought if Eddington’s experiment had not found the predicted bending of starlight. Einstein replied, “Then I would have been sorry for the dear Lord, for the theory is correct.”1

Of course, had experiments truly failed to confirm Einstein’s predictions, the theory would not be correct and general relativity would not have become a pillar of modern physics. But what Einstein meant is that general relativity describes gravity with such a deep inner elegance, with such simple yet powerful ideas, that he found it hard to imagine that nature could pass it by. General relativity, in Einstein’s view, was almost too beautiful to be wrong.

Aesthetic judgments do not arbitrate scientific discourse, however. Ultimately, theories are judged by how they fare when faced with cold, hard, experimental facts. But this last remark is subject to an immensely important qualification. While a theory is being constructed, its incomplete state of development often prevents its detailed experimental consequences from being assessed. Nevertheless, physicists must make choices and exercise judgments about the research direction in which to take their partially completed theory. Some of these decisions are dictated by internal logical consistency; we certainly require that any sensible theory avoid logical absurdities. Other decisions are guided by a sense of the qualitative experimental implications of one theoretical construct relative to another; we are generally not interested in a theory if it has no capacity to resemble anything we encounter in the world around us. But it is certainly the case that some decisions made by theoretical physicists are founded upon an aesthetic sense—a sense of which theories have an elegance and beauty of structure on par with the world we experience. Of course, nothing ensures that this strategy leads to truth.